

Customer Segmentation using RFM Analysis (BigQuery)

Project Overview

This project focuses on analyzing customer purchasing behavior using RFM (Recency, Frequency, Monetary) analysis. The objective is to segment customers based on their transaction patterns and identify high-value groups to support business decision-making, improve customer retention, and optimize marketing strategies.

Dataset Description

The dataset consists of transactional records containing customer identifiers, order dates, and purchase values. These attributes are essential for calculating RFM metrics. The data was processed and analyzed using Google BigQuery, which enables efficient handling of large-scale datasets.

Methodology

The analysis was conducted in multiple stages. First, data was aggregated to compute Recency (days since last purchase), Frequency (number of purchases), and Monetary (total spending). Next, scoring was applied using NTILE functions to assign relative rankings. Finally, customers were segmented into meaningful groups such as Champions, Loyal VIP, Promising, and At Risk based on their overall RFM scores.

RFM Metrics Output

```
1 select *
2 from rfm0708.sales.rfm_metrics
3 order by r_rank asc, f_rank asc, m_rank asc;
```

Query completed

Using on-demand processing quota

Query results

Job information	Results	Visualization	JSON	Execution details	Execution graph			
Row	customerid	last_order_date	recency	frequency	monetary	r_rank	f_rank	m_rank
1	CUST0049	2025-12-31	65	4	32.14	1	106	214
2	CUST0092	2025-12-31	65	5	113.27	2	74	31
3	CUST0012	2025-12-31	65	9	187.0900000000...	3	2	2
4	CUST0177	2025-12-31	65	5	122.91	4	72	24
5	CUST0294	2025-12-30	66	3	59.3799999999...	5	172	122
6	CUST0073	2025-12-30	66	5	142.38	6	68	9
7	CUST0236	2025-12-29	67	5	70.13	7	67	96
8	CUST0278	2025-12-29	67	2	27.89	8	213	224
9	CUST0213	2025-12-28	68	3	40.7399999999...	9	152	185
10	CUST0179	2025-12-27	69	5	55.54	10	66	138
11	CUST0034	2025-12-27	69	2	15.89	11	204	254
12	CUST0062	2025-12-27	69	3	51.0399999999...	12	184	157
13	CUST0277	2025-12-27	69	3	34.4800000000...	13	188	200
14	CUST0148	2025-12-27	69	5	90.62	14	60	56
15	CUST0215	2025-12-27	69	6	123.08	15	38	22

RFM Total Scores

1 select *

2 from rfm0708.sales.rfm_total_scores;

Query completed

Using on-demand processing quota

Query results

Job information

Results

Visualization

JSON

Execution details

Execution graph

Row	customerid	recency	frequency	monetary	r_score	f_score	m_score	rfm_total_score
1	CUST0012	65	9	187.0900000000...	10	10	10	30
2	CUST0289	73	7	121.1700000000...	10	10	10	30
3	CUST0227	73	6	126.89	10	10	10	30
4	CUST0264	73	6	126.84	10	10	10	30
5	CUST0215	69	6	123.08	10	9	10	29
6	CUST0226	74	6	103.7100000000...	10	10	9	29
7	CUST0084	82	8	123.51	9	10	10	29
8	CUST0017	73	6	90.0099999999...	10	10	8	28
9	CUST0039	88	8	167.4400000000...	8	10	10	28
10	CUST0104	74	6	164.74	9	9	10	28
11	CUST0102	71	5	108.9199999999...	10	9	9	28
12	CUST0073	66	5	142.38	10	8	10	28
13	CUST0267	88	6	140.89	8	10	10	28
14	CUST0262	79	5	121.5100000000...	9	9	10	28
15	CUST0157	85	7	132.83	8	10	10	28

Customer Segmentation Output

1 select *

2 from rfm0708.sales.rfm_segment_final;

Query completed

Using on-demand processing quota

Query results

Job information

Results

Visualization

JSON

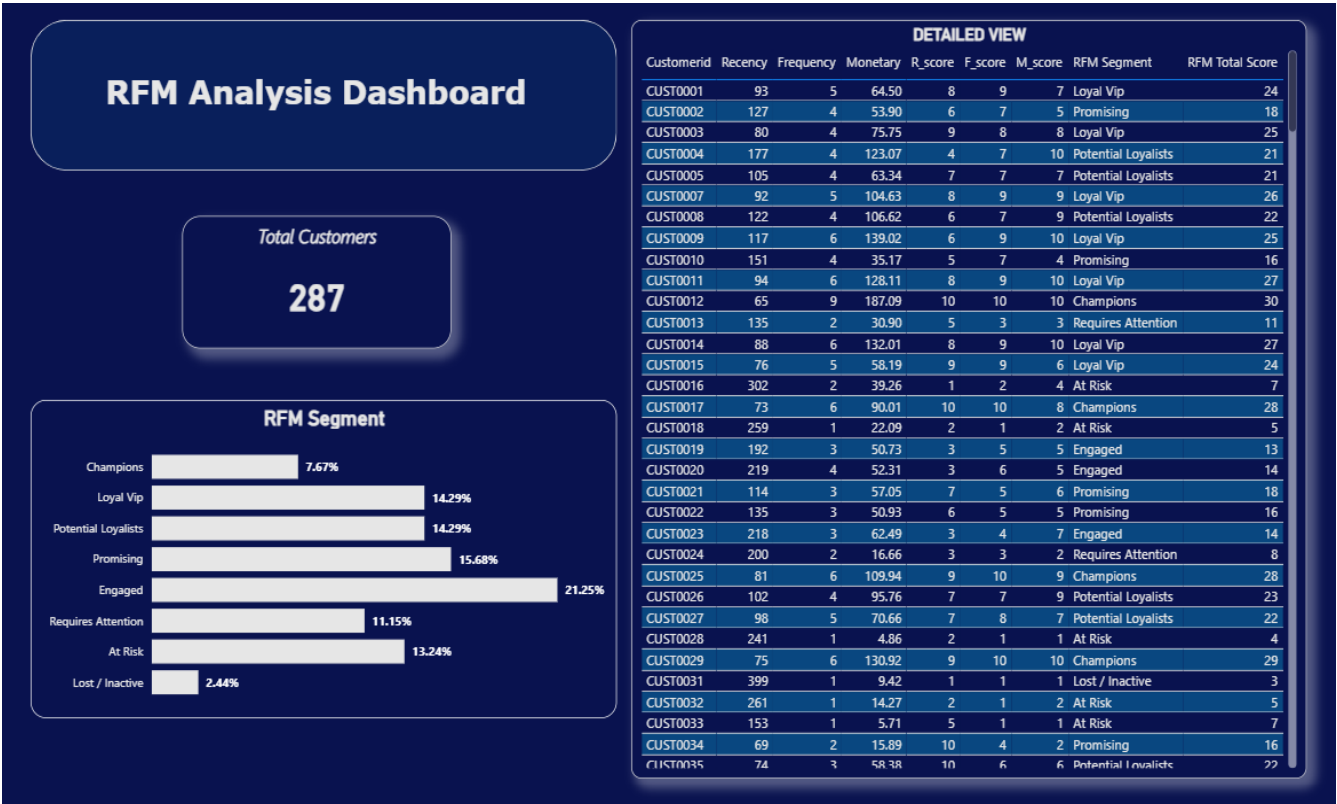
Execution details

Execution graph

Row	customerid	recency	frequency	monetary	r_score	f_score	m_score	rfm_total_score	rfm_segment
1	CUST0003	80	4	75.75	9	8	8	25	Loyal vip
2	CUST0115	225	3	52.66	3	4	5	12	Engaed
3	CUST0053	146	2	13.86	5	3	1	9	Requires Attention
4	CUST0045	207	2	23.2400000000...	3	3	2	8	Requires Attention
5	CUST0162	259	2	14.67	2	2	2	6	At Risk
6	CUST0057	236	1	24.63	3	1	2	6	At Risk
7	CUST0194	349	2	19.8100000000...	1	2	2	5	At Risk
8	CUST0011	94	6	128.1099999999...	8	9	10	27	Loyal vip
9	CUST0153	88	6	92.68	8	9	9	26	Loyal vip
10	CUST0233	107	3	61.04	7	5	6	18	Promising
11	CUST0002	127	4	53.9	6	7	5	18	Promising
12	CUST0125	123	3	41.3799999999...	6	5	4	15	Engaed
13	CUST0260	213	1	33.6	3	1	3	7	At Risk
14	CUST0178	75	5	114.12	9	9	9	27	Loyal vip
15	CUST0143	115	6	170.41	6	9	10	25	Loyal vip

RFM Dashboard (Power BI)

An interactive dashboard was developed using Power BI to visually represent customer segments and key metrics. The dashboard provides insights into total customers, distribution of segments, and detailed customer-level data. It enables stakeholders to quickly identify high-value segments and monitor customer behavior effectively.



Key Insights

The analysis reveals that Loyal VIP and Champions customers contribute significantly to revenue. A substantial portion of customers fall under Promising and Engaged categories, indicating potential for growth. Additionally, segments such as At Risk and Requires Attention highlight the need for targeted retention strategies.

Business Recommendations

- Offer exclusive rewards and loyalty programs to retain Champions and Loyal VIP customers.
- Implement targeted campaigns to re-engage At Risk and inactive customers.
- Nurture Promising customers through personalized offers and engagement strategies.
- Continuously monitor customer behavior using dashboards to support data-driven decisions.

Conclusion

This project demonstrates the practical application of RFM analysis using Google BigQuery and Power BI to derive actionable business insights from customer transaction data. By evaluating customers based on recency, frequency, and monetary value, the analysis successfully segments the customer base into meaningful categories such as Champions, Loyal VIP, Promising, and At Risk. These segments enable businesses to tailor their marketing strategies and focus on high-impact areas. The integration of BigQuery allowed efficient processing of large-scale data, while Power BI facilitated intuitive visualization of customer segments and performance metrics. The dashboard provides a clear and interactive view of customer behavior, supporting faster and more informed decision-making. Overall, this approach helps organizations improve customer retention, increase engagement, and maximize revenue by targeting the right customers with the right strategies. The project highlights the importance of combining data engineering, analytical techniques, and visualization tools to deliver end-to-end business intelligence solutions.